



Specification

Specification Book

Scope

Applicable Document of Part 1
Common Electrical Specification for Supplier

Basic DID Definition Specification

Range Of Validity

VinFast Vehicles

Author: Daniel Canelopol

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Specification

Approval Confirmed By

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1 General Information

1.1 Language

This specification is written in English.

1.2 Change History

Version	Chapter	Brief Description	Date	Edited By
1.0.0	All	# Initial Version	2019.10.14	Nguyen Hoang Long
1.0.0	All	# Pdf -> Word Conversion and further update w.r.to C-SUV	2020.01.02	Rasul Nadaf
1.0.1	All	# Change DID 0xF18A from 5 byte ASCII to 4 byte BCD # Changed usage of DID 0xF190 as Mandatory for EMS, BCM & MHU # Changed usage of DID 0xF102 (was previously mandatory) # Changed parameter example of DID 0xF103, - 7th Byte low nibble is needed - 8th byte marked as ASCII Character "V" to match with Team center # Changed usage of DID 0xF111 marked for BCM/XGW and MHU # Changed usage of DID 0xF112 marked for BCM/XGW and MHU # Changed usage of DID 0xF113 marked for BCM/XGW and MHU # Changed DID 0xF11A name from PT to Powertrain # Changed DID 0xF11B name from TC to Chassis # Deleted DID 0xF11F as its not applicable # Changed usage of DID 0xFD01 (was previously mandatory) # Changed usage of DID 0xF194(was previously mandatory) # Added below New DID as per VF35 architecture • DID 0xF125 ECU ADAS CAN dbc file version • DID 0xF126 ECU Diagnostic CAN dbc file version • DID 0xF127 ECU FA CAN dbc file version • DID 0xF128 ECU FLEXRAY FIBEX file version • DID 0xF129 ECU LIN3 LDF file version	2021.04.25	Rasul Nadaf
1.1	3	1) Specification document name changed to VFDSVAEEP0002. 3) Added below New DID as per architecture: • DID 0xF120 Time Stamp - Global snapshot • DID 0xF121 Odometer - Global snapshot • DID 0xF122 ECU Supply Voltage - Global snapshot • DID 0xF123 Ignition status - Global snapshot • DID 0xF18C ECUSerialNumberDataIdentifier	2021.10.22	Daniel Canelopol

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		- DID DD01 Cabin Temperature - DID DD02 Outside Temperature - DID DD03 ECU Temperature - DID DD04 ECU Voltage - DID D00D Vehicle Communication Bus State - DID DD06 Engine status - Global snapshot - DID DD07 GPS Time - Global snapshot - DID DD08 Engine RPM - Global snapshot - DID DD09 Vehicle speed - Global snapshot - DID F18F RegulationXSoftware Identification Number (RxSWIN) - DID D020 Internal Fail Code - DID 4884 Internal Fault Data - DID F800 VIN2 - DID F801 VIN3 - DID F802 VIN4 - DID F141 Hardware Number Revision Level - DID F142 Calibration Software Number Revision Level - DID F143 Purchase Part Number Revision Level - DID F144 Application Software Part Number Revision Level - DID F145 Basic Software Part Number Revision Level - DID F146 HMI Software Part Number Revision Level - DID F148 Software Part Number Revision Level 4) Removed DID F18A.		Daniel Canelopol
1.2	3	- Format of the following DID's has been changed to match Team Center Part Number – F188, F191, F102, F103, F104, F105, F106. - Flash Boot Loader defined for all DID's. - CAN signals referenced for Snapshot DID's	2021.11.29	Daniel Canelopol
1.3	3	- F18C DID – Write once in Bootloader added - D005 DID changed to DD05 - F120 DID name change to Global Time - F18F DID – parameter details added - D020 DID – parameter details updated 016C DID – new DID added - D05B DID – new DID added - D06B DID – new DID added	2022.01.28	Daniel Canelopol

Table 1: Change History

2 Introduction

2.1 Coherence to other documents

The “Generic_Diagnostic_Specification” refers to this document.

Specification

2.2 Terms and definitions

The following list of terms and definitions are used throughout this document with the given meaning.

2.2.1 Abbreviations

Abbreviation	meaning
ECU	Electronic Control Unit
DID	Data Identifier
PID	Parameter Identifier
BOOT	Bootloader
Cvt	Convention
C	Conditional
M	Mandatory
O	Optional
U	User Conditional

Table 2 : List of abbreviations Definitions

2.2.1.1 Onligation of Requirement Wording

The following wording rules are used throughout this document.

Shall

The word "shall" is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

Should

The word "should" is used for preferences or desires in implementation that are not strictly binding.

May

The word "may" is used to denote options that are left to the implementer.

2.3 Security Level

The number and usage of Security Levels is described in the Generic_Diagnostic_Specification. Basically, the following Security Level exists:

Level	Request Seed	Send Key	Use case
1	0x01	0x02	Service
2	0x03	0x04	Manufacturing
3	0x05	0x06	Variant Coding
4	0x07	0x08	Flashing (ECU reprogramming)

Table 3 : Security Level

Level 0 means no security access is required.

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3 System Description

3.1 DID 0xF186 Active Diagnostic Session

Service	0x22																															
DID	0xF186																															
Short Description	Read Active Diagnostic Session																															
Implementation	Application		Flash Boot Loader																													
	Mandatory Read only		Read																													
Security Level	Read		Write																													
	Level 0		N/A																													
Usage	Mandatory for all CAN ECUs and diagnostic Class III LIN node.																															
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td>4</td><td>sub-function = [DiagnosticSessionType</td><td></td><td>00 – FF</td></tr><tr><td></td><td>defaultSession</td><td>M</td><td>01</td></tr><tr><td></td><td>programmingSession</td><td>M</td><td>02</td></tr><tr><td></td><td>extendedDiagnosticSession</td><td>M</td><td>03</td></tr><tr><td></td><td>coding session</td><td>M</td><td>41</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value	4	sub-function = [DiagnosticSessionType		00 – FF		defaultSession	M	01		programmingSession	M	02		extendedDiagnosticSession	M	03		coding session	M	41		]		
	Data Byte No.	Parameter Name	Cvt	Value																												
	4	sub-function = [DiagnosticSessionType		00 – FF																												
		defaultSession	M	01																												
		programmingSession	M	02																												
		extendedDiagnosticSession	M	03																												
		coding session	M	41																												
		]																														
	Parameter:																															
	- Active Diagnostic Session: 01 = Default Session 02 = Programming Session 03 = Extended Diagnostic Session 41 = Coding Session																															
	Note:																															
	These are the sub function parameters from Diagnostic Session Control service.																															
See also Generic_Diagnostic_Specification chapter “Diagnostic Services”.																																

Specification

3.2 DID 0xF188 Vehicle Manufacturer ECU Software Part Number

Service	0x22																	
DID	0xF188																	
Short Description	Software Part Number of the ECU defined by the OEM.																	
Implementation	Application	Flash Boot Loader																
	Mandatory Read(physical and functional addressing)	Updated during software download Not readable in Boot Loader																
Security Level	Read	Write																
	Level 0	N/A																
Usage	Mandatory for all ECUs																	
Data	Data Byte No.	Parameter Name	Cvt	Value														
		ECU Software Part Number [] = [																
	4	ECU Software Number Prefix Byte#0	M	ASCII														
	5	ECU Software Number Prefix Byte#1	M	ASCII														
	6	ECU Software Number Prefix Byte#2	M	ASCII														
	7	ECU Software Number Byte#0	M	BCD														
	8	ECU Software Number Byte#1	M	BCD														
	9	ECU Software Number Byte#2	M	BCD														
	10	ECU Software Number Byte#3	M	BCD														
	Parameter:																	
	ECU Software Part Number from BOM/Teamcenter: e.g.: SOW12345678																	
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	53	4F	57	12	34	56	78
	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10											
53	4F	57	12	34	56	78												
Note:																		
- The OEM defines Vehicle Manufacture ECU Software Part Number, and it shall be embedded in the ECU Software file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Vehicle Manufacture ECU Software Part Number is stored. - Also see DID 0xF148 Vehicle Manufacturer ECU Software Part Number Revision Level.																		

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3.3 DID 0xF18F RegulationXSoftware Identification Number (RxSWIN)

Service	0x22			
DID	0xF18F			
Short Description	RegulationXSoftware Identification Number (RxSWIN)			
Implementation	Application		Flash Boot Loader	
	Mandatory Read(physical and functional addressing)		Updated during software download Not readable in PBL	
Security Level	Read		Write	
	Level 0		N/A	
Usage	Mandatory for ECUs that must adhere to RegulationXSWin			
Data	Data Byte No.	Parameter Name	Cvt	Value
		RegulationXSoftware Identification Number [] = [		
	4	RegulationXSoftware Identification Number HighByte#0 [] = [	M	ASCII
	36	RegulationXSoftware Identification Number HighByte#31[] = [		
		]		
Parameter: • The values should be aligned left. • Unused Bytes should be filled with space (0x20) • RxSWIN Identification number is a dedicated identifier, representing informations about the type approval relevant software of the Electronic Control System contributing to the Regulation Number X type approval relevant characteristics of the vehicle. • Please refer to ECE-R156 & ISO14229-1 Annex C.				
Note: • The ECU name shall be defined by the OEM, but should be written by the Supplier.				

3.4 DID 0xF18B ECU Manufacturing Date

Service	0x22	
DID	0xF18B	
Short Description	Read ECU Manufacturing Date	
Implementation	Application	Flash Boot Loader
	Mandatory Read(physical and functional addressing)	N/A
Security Level	Read	Write
	Level 0	N/A
Usage	Mandatory for all ECUs	
Data		

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Data Byte No.	Parameter Name	Cvt	Value
	ECU Manufacturing Date[] = [		
4	ECU Manufacturing Date - Year	M	(00 - 99) BCD
5	ECU Manufacturing Date - Month	M	(01 - 12) BCD
6	ECU Manufacturing Date - Day	M	(01 - 31) BCD
	]		

Parameter:

- Example ECU Manufacturing Date format:
e.g: 2013.01.21

Byte4	Byte5	Byte6
13	01	21

Note:

- Writing the date shall be done at the supplier site.
- The ECU Manufacturing Date shows the production date of the ECU.
- All values 00 means no ECU Manufacturing Date is stored.

3.5 DID 0xF190 Vehicle Identification Number (VIN)

Service	0x22 /0x2E																											
DID	0xF190																											
Short Description	Read / Write complete Vehicle Identification Number (17 Characters)																											
Implementation	Application	Flash Boot Loader																										
	Mandatory Read/Write	Read																										
Security Level	Read	Write																										
	Level 0	Level1&2																										
Usage	Mandatory for all ECU's that supports diagnostics																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>4</td><td>Vehicle Identification Number High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>20</td><td>Vehicle Identification Number Low Byte#16</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			4	Vehicle Identification Number High Byte#0	M	ASCII	...				20	Vehicle Identification Number Low Byte#16	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	4	Vehicle Identification Number High Byte#0	M	ASCII																								
	...																											
	20	Vehicle Identification Number Low Byte#16	M	ASCII																								
		]																										
	Parameter:																											
	- Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																											
	Note:																											
- The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later processes. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER																												

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3.6 DID 0xF191 Vehicle Manufacturer ECU Hardware Part Number

Service	0x22																
DID	0xF191																
Short Description	Hardware Part Number of the ECU defined by the OEM.																
Implementation	Application	Flash Boot Loader															
	Mandatory Read(physical and functional addressing)	Read															
Security Level	Read	Write															
	Level 0	N/A															
Usage	Mandatory																
Data	Data Byte No.	Parameter Name	Cvt	Value													
		ECU Hardware Part Number [] = [															
	4	ECU Hardware Number Prefix Byte#0	M	ASCII													
	5	ECU Hardware Number Prefix Byte#1	M	ASCII													
	6	ECU Hardware Number Prefix Byte#2	M	ASCII													
	7	ECU Hardware Number Byte#0	M	BCD													
	8	ECU Hardware Number Byte#1	M	BCD													
	9	ECU Hardware Number Byte#2	M	BCD													
	10	ECU Hardware Number Byte#3	M	BCD													
		]															
	Parameter:																
	ECU Hardware Part Number from BOM/Teamcenter:																
	e.g.: EEP12345678																
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>45</td><td>45</td><td>50</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	45	45	50	12	34	56
Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10											
45	45	50	12	34	56	78											

Note:

- The OEM defines Vehicle Manufacture ECU Hardware Part Number.
- The Hardware Part Number must be written by the Supplier.
- This DID is used to identify a unique combination of ECU Hardware and non-reprogrammable software (e.g. Boot Software).
- If ECU hardware or the Boot Software Version is changed in the ECU delivered to VINFAST, then the Vehicle Manufacturer ECU Hardware Part Number or Revision Level must be changed.
- All Bytes 0x00 means no Vehicle Manufacture ECU Hardware Part Number is stored.
- Also see DID 0xF141 Vehicle Manufacturer ECU Hardware Part Number Revision Level.

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3.7 DID 0XF102 Vehicle Manufacturer ECU Calibration Part Number

Service	0x22																	
DID	0xF102																	
Short Description	Calibration Software Number of the ECU defined by the OEM																	
Implementation	Application	Flash Boot Loader																
	Mandatory Read	Updated during software download Not readable in Boot Loader																
Security Level	Read	Write																
	Level 0	N/A																
Usage	Mandatory for ECUs where Calibration Part Number is to be reported separately																	
	Data Byte No.	Parameter Name	Cvt	Value														
		Calibration Software Part Number [] = [																
	4	ECU Calibration Number Prefix Byte#0	M	ASCII														
	5	ECU Calibration Number Prefix Byte#1	M	ASCII														
	6	ECU Calibration Number Prefix Byte#2	M	ASCII														
	7	ECU Calibration Number Byte#0	M	BCD														
	8	ECU Calibration Number Byte#1	M	BCD														
	9	ECU Calibration Number Byte#2	M	BCD														
	10	ECU Calibration Number Byte#3	M	BCD														
	Parameter:																	
	ECU Calibration Part Number from BOM/Teamcenter: e.g.: SOW11223344																	
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>11</td><td>22</td><td>33</td><td>44</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	53	4F	57	11	22	33	44
	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10											
	53	4F	57	11	22	33	44											
	Note:																	
	- The Calibration Part Number is defined by the OEM, and shall be embedded in the ECU Calibration file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Vehicle Manufacture ECU Calibration Part Number is stored. - Also see DID 0xF142 Vehicle Manufacturer ECU Calibration Part Number Revision Level.																	

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3.8 DID 0XF101 Boot Loader Version Number

Service	0x22																										
DID	0xF101																										
Short Description	Read Boot Loader Version number																										
Implementation	Application	Flash Boot Loader																									
	Read	Read																									
Security Level	Read	Write																									
	Level 0	N/A																									
Usage	Mandatory for the ECU's with flash functionality																										
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Boot loader version number [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU Boot loader version number high byte#0- - High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU Boot loader version number high byte#0- Low nibble</td><td>M</td><td>(00-99) BCD</td></tr><tr><td>5</td><td>ECU Boot loader version number high byte#1</td><td>M</td><td>(00-99) BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>			Data Byte No.	Parameter Name	Cvt	Value		ECU Boot loader version number [] = [			4	ECU Boot loader version number high byte#0- - High nibble Byte	M	Unsigned	4	ECU Boot loader version number high byte#0- Low nibble	M	(00-99) BCD	5	ECU Boot loader version number high byte#1	M	(00-99) BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																							
		ECU Boot loader version number [] = [																									
	4	ECU Boot loader version number high byte#0- - High nibble Byte	M	Unsigned																							
	4	ECU Boot loader version number high byte#0- Low nibble	M	(00-99) BCD																							
	5	ECU Boot loader version number high byte#1	M	(00-99) BCD																							
		]																									
	Parameter: Example Boot loader version number: e.g: V12.34																										
	<table><tr><td>Byte4 High Nibble</td><td>Byte 4 Low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>			Byte4 High Nibble	Byte 4 Low Nibble	Byte5	1	2	34																		
	Byte4 High Nibble	Byte 4 Low Nibble	Byte5																								
	1	2	34																								
	Note: - This identifier shall be defined by supplier and written at the supplier side, and shall not be able to be modified during the vehicle life. - All values 0x00 means no System Supplier Boot Loader Version Number is stored.																										

Specification

3.9 DID 0xF103 Vehicle Manufacturer ECU Purchase Part Number

Service	0x22																
DID	0xF103																
Short Description	Part Number of the ECU defined by the OEM																
Implementation	Application	Flash Boot Loader															
	Mandatory Read (physical and functional addressing)	Updated during software download Not readable in Boot Loader															
Security Level	Read	Write															
	Level 0	N/A															
Usage	Mandatory																
Data	Data Byte No.	Parameter Name	Cvt	Value													
		ECU Part Number [] = [															
	4	ECU Part Number Prefix Byte#0	M	ASCII													
	5	ECU Part Number Prefix Byte#1	M	ASCII													
	6	ECU Part Number Prefix Byte#2	M	ASCII													
	7	ECU Part Number Byte#0	M	BCD													
	8	ECU Part Number Byte#1	M	BCD													
	9	ECU Part Number Byte#2	M	BCD													
	10	ECU Part Number Byte#3	M	BCD													
	Parameter:																
	ECU Purchase Part Number from BOM/Team center:																
	e.g. EEP12345678																
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>45</td><td>45</td><td>50</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	45	45	50	12	34	56
Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10											
45	45	50	12	34	56	78											

Note:						
- The ECU Purchase Part Number is defined by the OEM and must be written at Supplier manufacturing process. - This DID is used to identify a unique combination of ECU Hardware and reprogrammable Software and Calibration. - If ECU Hardware or Software or Calibration are changed in the ECU delivered to VINFAST, then the Vehicle Manufacturer ECU Purchase Part Number or Revision Level <u>must be changed</u> - All Bytes 0x00 means no ECU Purchase Part Number is stored. - Also see DID 0xF143 Vehicle Manufacturer ECU Purchase Part Number Revision Level						

Specification

3.10 DID 0xF104 Vehicle Manufacturer ECU Application Software Part Number

Service	0x22																	
DID	0xF104																	
Short Description	Application Software Part Number defined by the OEM																	
Implementation	Application		Flash Boot Loader															
	Read		Updated during software download Not readable in PBL															
Security Level	Read		Write															
	Level 0		N/A															
Usage	Mandatory for ECUs with flashing function and whose application software (ASW) packages can be programmed separately from basic software (BSW), else use DID 0xF188																	
Data	Data Byte No.	Parameter Name	Cvt	Value														
		ECU Application Software Part Number []																
		= [																
	4	ECU Application Software Number Prefix Byte#0	M	ASCII														
	5	ECU Application Software Number Prefix Byte#1	M	ASCII														
	6	ECU Application Software Number Prefix Byte#2	M	ASCII														
	7	ECU Application Software Number Byte#0	M	BCD														
	8	ECU Application Software Number Byte#1	M	BCD														
	9	ECU Application Software Number Byte#2	M	BCD														
	10	ECU Application Software Number Byte#3	M	BCD														
		]																
	Parameter:																	
	Application Software Part Number from BOM/Teamcenter e.g. SOW12345678 <table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	53	4F	57	12	34	56	78
	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10											
53	4F	57	12	34	56	78												
Note:																		
- The Application Software Number is defined by the OEM, and shall be embedded in the ECU Application Software file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Application Software Number is stored. - Also see DID 0xF144 Vehicle Manufacturer ECU Application Software Part Number Revision Level																		

Specification

3.11 DID 0xF105 Vehicle Manufacturer ECU Basic Software Part Number

Service	0x22																	
DID	0xF105																	
Short Description	Basic Software Part Number defined by the OEM																	
Implementation	Application	Flash Boot Loader																
	Read	Updated during software download Not readable in PBL																
Security Level	Read	Write																
	Level 0	N/A																
Usage	Mandatory for ECUs with flashing function and whose basic software (BSW) packages can be programmed separately from application software (ASW), else use DID 0xF188.																	
Data	Data Byte No.	Parameter Name	Cvt	Value														
		ECU Basic Software Part Number [] = []																
	4	ECU Basic Software Number Prefix Byte#0	M	ASCII														
	5	ECU Basic Software Number Prefix Byte#1	M	ASCII														
	6	ECU Basic Software Number Prefix Byte#2	M	ASCII														
	7	ECU Basic Software Number Byte#0	M	BCD														
	8	ECU Basic Software Number Byte#1	M	BCD														
	9	ECU Basic Software Number Byte#2	M	BCD														
	10	ECU Basic Software Number Byte#3	M	BCD														
	Parameter: Basic Software Part Number from BOM/Teamcenter e.g. SOW12345678 <table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table>					Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	53	4F	57	12	34	56
Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10												
53	4F	57	12	34	56	78												
Note: - The Basic Software Part Number is defined by the OEM, and shall be embedded in the ECU Basic Software file. - This DID is updated during the reprogramming process. - All Bytes 0x00 means no Basic Software Part Number is stored. - Also see DID 0xF145 Vehicle Manufacturer ECU Basic Software Part Number Revision Level																		

Specification

3.12 DID 0xF106 Vehicle Manufacturer ECU HMI Software Data Part Number

Service	0x22																	
DID	0xF106																	
Short Description	HMI Software Part Number defined by the OEM																	
Implementation	Application		Flash Boot Loader															
	Read		Updated during software download Not readable in PBL															
Security Level	Read		Write															
	Level 0		N/A															
Usage	Mandatory for ECUs with HMI software module and can be reprogrammed.																	
Data	Data Byte No.	Parameter Name	Cvt	Value														
		ECU HMI Software Part Number [] =																
	4	ECU HMI Software Number Prefix Byte#0	M	ASCII														
	5	ECU HMI Software Number Prefix Byte#1	M	ASCII														
	6	ECU HMI Software Number Prefix Byte#2	M	ASCII														
	7	ECU HMI Software Number Byte#0	M	BCD														
	8	ECU HMI Software Number Byte#1	M	BCD														
	9	ECU HMI Software Number Byte#2	M	BCD														
	10	ECU HMI Software Number Byte#2	M	BCD														
	Parameter:																	
	HMI Software Part Number from TeamCenter e.g. SOW12345678																	
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td><td>Byte10</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	53	4F	57	12	34	56	78
	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10											
53	4F	57	12	34	56	78												
Note:																		
- This HMI Software Part Number is defined by OEM, and shall be embedded in the ECU HMI file. - This DID will be updated during the reprogramming process. - All Bytes 0x00 means no HMI Software Number is stored - Also see DID 0xF146 Vehicle Manufacturer ECU HMI Software Data Part Number Revision Level																		

Specification

3.13 DID 0xF107 Programming Fingerprint

Service	0x22 / 0x2E																																							
DID	0xF107																																							
Short Description	Programming fingerprint																																							
Implementation	Application		Flash Boot Loader																																					
	Read		Manadatory Write for Flashable ECU's																																					
Security Level	Read		Write																																					
	Level 0		Level 4(programming session)																																					
Usage	Mandatory for the ECUs with flash functionality																																							
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Write Fingerprint[] = [</td><td></td><td></td></tr><tr><td>4</td><td>programmingDate [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr><tr><td>5</td><td>programmingDate [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr><tr><td>6</td><td>programmingDate [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr><tr><td>7</td><td>repairShopCodeOrTesterSerialNumber[byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td></td><td></td></tr><tr><td>33</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Write Fingerprint[] = [			4	programmingDate [byte#1] (YY)	M	BCD(00-99)	5	programmingDate [byte#2] (MM)	M	BCD(01-12)	6	programmingDate [byte#3] (DD)	M	BCD(01-31)	7	repairShopCodeOrTesterSerialNumber[byte#1] (MSB)	M	ASCII	...				33	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																																				
		Write Fingerprint[] = [																																						
	4	programmingDate [byte#1] (YY)	M	BCD(00-99)																																				
	5	programmingDate [byte#2] (MM)	M	BCD(01-12)																																				
	6	programmingDate [byte#3] (DD)	M	BCD(01-31)																																				
	7	repairShopCodeOrTesterSerialNumber[byte#1] (MSB)	M	ASCII																																				
	...																																							
	33	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII																																				
		]																																						
	Parameter: e.g.: 130128 2W...KL																																							
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>...</td><td>Byte 32</td><td>Byte33</td></tr><tr><td>13</td><td>01</td><td>28</td><td>2</td><td>W</td><td>...</td><td>K</td><td>L</td></tr></table>				Byte4	Byte5	Byte6	Byte7	Byte8	...	Byte 32	Byte33	13	01	28	2	W	...	K	L																				
	Byte4	Byte5	Byte6	Byte7	Byte8	...	Byte 32	Byte33																																
	13	01	28	2	W	...	K	L																																
	Note: - Defined by OEM and written in aftersale. - Programming Date is BCD code, and default value is 0x00. - Byte7 to Byte16 represents the Repair Shop Code in ASCII format. - the following 17 bytes represent the tester Serial number, and the tester serial number is reserved. - default value is all bytes space(0x20).																																							
	0xF131 to 0xF135 are the blocked and used to store the programming fingerprint. Programming fingerprint written should use DID 0xF107 by the service 0x2E. The data shall be stored at 0xF131 if the fingerprint data is written for the first time. If another data need to be written (also with 0xF107), block 0xF132 should be filled and so on. Reading 0xF107 shall always return that last programming fingerprint that was written. After the last block of 0xF135 is filled and then if all available block are populated, 0xF132 shall be overwritten, then 0xF133 and so on. 0xF131 should be leaved and can't be overwritten. DID F131 to F135 shall be readable as same as DID F107 including the length and format.																																							

Specification

3.14 DID 0xF111 Factory Vehicle Order Data VOD Data Identifier

Service	0x22 /0x2E		
DID	0xF111		
Short Description	Read / Write factory Vehicle Order Data VOD		
Implementation	Application	Flash Boot Loader	
	Optional Read / Write	N/A	
Security Level	Read	Write	
	Level 0	Level 3	
Usage	Mandatory for any ECU category 1 & 2		
Data			
	Data Byte No.	Parameter Name	Cvt Value
		Factory Vehicle Order Data VOD [] = [	
	4	Factory Vehicle Order Data VOD Format Byte	M Integer
	5	Factory Vehicle Order Data VOD Data High Byte#0	M Mixed
	...		
	n	Factory Vehicle Order Data VOD Data Low Byte#n	M Mixed
	n+1	Factory Vehicle Order Data VOD CRC	M Integer
		]	

Specification

3.15 DID 0xF112 Current Vehicle Order Data VOD Data Identifier

Service	0x22 /0x2E																																			
DID	0xF112																																			
Short Description	Read / Write Current Vehicle Order Data VOD																																			
Implementation	Application	Flash Boot Loader																																		
	Optional Read / Write	N/A																																		
Security Level	Read	Write																																		
	Level 0	Level 3																																		
Usage	Mandatory for any ECU category 1 & 2																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Current Vehicle Order Data VOD [] = [</td><td></td><td></td></tr><tr><td>4</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr><tr><td>5</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>...</td></tr><tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Current Vehicle Order Data VOD [] = [			4	Current Vehicle Order Data VOD Format Byte	M	Integer	5	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed	...		...	...	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed	n+1	Current Vehicle Order Data VOD CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																
		Current Vehicle Order Data VOD [] = [																																		
	4	Current Vehicle Order Data VOD Format Byte	M	Integer																																
	5	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed																																
	...		...	...																																
	n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed																																
	n+1	Current Vehicle Order Data VOD CRC	M	Integer																																
		]																																		
	Parameter:																																			
	Example Vehicle Order Data VOD:																																			
	- Format Byte = 00000101 = 5dec - Length of the Vehicle Order Data VOD Data = 5 Byte - The content of this DID is defined and managed by the Project. - The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD																																			
	Note:																																			
	- DID would be defined and written by OEM and ECU supplier together. - All Bytes 0 means no Factory Vehicle Order Data VOD is stored. - The Vehicle Order Data VOD includes the Format Byte, the n Byte field, which represents the description of the options and topology of the whole vehicle as bit pattern and the CRC in the last Byte. The Vehicle Order Data VOD is written in two ECUs in the vehicle. - Current VOD shall be same as factory VOD if function/features/options of the whole vehicle is never been changed. - Service shall update this Current VOD should a feature/function/option change in service that is impacted by the coding strategy. - For more information please refer the Coding Specification and VOD definition sheet.																																			

Specification

3.16 DID 0xF108 Coding Data String CDS Data Identifier

Service	0x22 /0x2E																																																			
DID	0xF108																																																			
Short Description	Read / Write Coding Data String CDS																																																			
Implementation	Application	Flash Boot Loader																																																		
	Read / Write	N/A																																																		
Security Level	Read	Write																																																		
	Level 0	Level 3																																																		
Usage	Mandatory for the ECUs with Coding functionality																																																			
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Coding Data String CDS[] = [</td><td></td><td></td></tr><tr><td>4</td><td>CDS Format Byte High Byte#0</td><td>M</td><td>Integer</td></tr><tr><td>5</td><td>CDS Format Byte Low Byte#1</td><td>M</td><td>Integer</td></tr><tr><td>6</td><td>CDS Short VIN High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>....</td><td>...</td><td>....</td></tr><tr><td>12</td><td>CDS Short VIN Low Byte#6</td><td>M</td><td>ASCII</td></tr><tr><td>13</td><td>CDS Data High Byte#0</td><td>M</td><td>Mixed</td></tr><tr><td>...</td><td>....</td><td>...</td><td>....</td></tr><tr><td>n</td><td>CDS Data Low Byte#n</td><td>M</td><td>Mixed</td></tr><tr><td>n+1</td><td>CDS CRC</td><td>M</td><td>Integer</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Coding Data String CDS[] = [			4	CDS Format Byte High Byte#0	M	Integer	5	CDS Format Byte Low Byte#1	M	Integer	6	CDS Short VIN High Byte#0	M	ASCII	...		...		12	CDS Short VIN Low Byte#6	M	ASCII	13	CDS Data High Byte#0	M	Mixed	...		...		n	CDS Data Low Byte#n	M	Mixed	n+1	CDS CRC	M	Integer		]		
	Data Byte No.	Parameter Name	Cvt	Value																																																
		Coding Data String CDS[] = [																																																		
	4	CDS Format Byte High Byte#0	M	Integer																																																
	5	CDS Format Byte Low Byte#1	M	Integer																																																
	6	CDS Short VIN High Byte#0	M	ASCII																																																
	...		...																																																	
	12	CDS Short VIN Low Byte#6	M	ASCII																																																
	13	CDS Data High Byte#0	M	Mixed																																																
	...		...																																																	
	n	CDS Data Low Byte#n	M	Mixed																																																
	n+1	CDS CRC	M	Integer																																																
		]																																																		
	Parameter:																																																			
	Example Coding Data String CDS:																																																			
	- Format Byte = 00 0010 1000 1100 = 652 dec - Length of the Coding Data String = 652 Bytes																																																			
	Note:																																																			
	- This data would be defined by OEM together with supplier,and shall be written at OEM site. - All Bytes 0 means no Coding Data String is stored. - The Coding Data String includes 10bits CDS format, - A Short VIN (last 7 characters of the 0xF190 Vehicle Identification Number) - The n Bytes field, which represent the Coding information as bit pattern and the CRC in the last Byte. - The Coding Data String field is limited to 1024 Bytes. - For more information, please refer the Coding Specification.																																																			

Specification

3.17 DID 0xF10A Coding for CDS Fingerprint

Service	0x22 /0x2E																																							
DID	0xF10A																																							
Short Description	Coding for CDS fingerprint																																							
Implementation	Application	Flash Boot Loader																																						
	Read / Write	N/A																																						
Security Level	Read	Write																																						
	Level 0	Level 3																																						
Usage	Mandatory for the ECUs with Coding functionality.																																							
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Coding for CDS fingerprint[] = [</td><td></td><td></td></tr><tr><td>4</td><td>codingDate [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr><tr><td>5</td><td>codingDate [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr><tr><td>6</td><td>codingDate [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr><tr><td>7</td><td>repairShopCodeOrTesterSerialNumber [byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>....</td><td>...</td><td>....</td></tr><tr><td>33</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Coding for CDS fingerprint[] = [			4	codingDate [byte#1] (YY)	M	BCD(00-99)	5	codingDate [byte#2] (MM)	M	BCD(01-12)	6	codingDate [byte#3] (DD)	M	BCD(01-31)	7	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII	...		...		33	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																																				
		Coding for CDS fingerprint[] = [																																						
	4	codingDate [byte#1] (YY)	M	BCD(00-99)																																				
	5	codingDate [byte#2] (MM)	M	BCD(01-12)																																				
	6	codingDate [byte#3] (DD)	M	BCD(01-31)																																				
	7	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII																																				
	...		...																																					
	33	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII																																				
		]																																						
	Parameter: Coding for CDS fingerprint format: 130128xxxxxxxxxx																																							
	Note: - Defined and written by OEM. - All byte 0 means no CDS fingerprint is stored. - Coding Date is BCD code. - Byte 7 to Byte16 represents the Repair Shop Code in ASCII format. - The following17 bytes represent the Tester Serial number, and the Tester serial number is reserved.																																							

Specification

3.18 DID 0xF113 Coding for VOD Fingerprint

Service	0x22 /0x2E																																							
DID	0xF113																																							
Short Description	Coding for VOD fingerprint																																							
Implementation	Application	Flash Boot Loader																																						
	Read / Write	N/A																																						
Security Level	Read	Write																																						
	Level 0	Level 3																																						
Usage	Mandatory for any ECU category 1 & 2																																							
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Coding for VOD fingerprint[] = [</td><td></td><td></td></tr><tr><td>4</td><td>codingDate [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr><tr><td>5</td><td>codingDate [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr><tr><td>6</td><td>codingDate [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr><tr><td>7</td><td>repairShopCodeOrTesterSerialNumber [byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>....</td><td>...</td><td>....</td></tr><tr><td>33</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Coding for VOD fingerprint[] = [			4	codingDate [byte#1] (YY)	M	BCD(00-99)	5	codingDate [byte#2] (MM)	M	BCD(01-12)	6	codingDate [byte#3] (DD)	M	BCD(01-31)	7	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII	...		...		33	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																																				
		Coding for VOD fingerprint[] = [																																						
	4	codingDate [byte#1] (YY)	M	BCD(00-99)																																				
	5	codingDate [byte#2] (MM)	M	BCD(01-12)																																				
	6	codingDate [byte#3] (DD)	M	BCD(01-31)																																				
	7	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII																																				
	...		...																																					
	33	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII																																				
		]																																						
	Parameter: Coding for VOD fingerprint format: 130128xxxxxxxxxx																																							
	Note: - Defined and written by OEM. - All byte 0 means no VOD fingerprint is stored. - Coding Date is BCD code. - Byte 7 to Byte16 represents the Repair Shop Code in ASCII format. - The following17 bytes represent the Tester Serial number, and the Tester serial number is reserved.																																							

Specification

3.19 DID 0xF10C ECU Body CAN dbc file version.

Service	0x22																											
DID	0xF10C																											
Short Description	ECU body CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all body CAN ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU body CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU body CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU body CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU body CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU body CAN .dbc version [] = [			4	ECU body CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	4	ECU body CAN .dbc version -high byte#0- low nibble Byte	M	BCD	5	ECU body CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU body CAN .dbc version [] = [																										
	4	ECU body CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	4	ECU body CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	5	ECU body CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 1234																											
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																									
	1	2	34																									
Note:																												
- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU DBC file version is stored. - All ECU's connected on Body CAN should support this DID.																												

Specification

3.20 DID 0xF11A ECU Powertrain CAN dbc file version

Service	0x22																											
DID	0xF11A																											
Short Description	ECU PT CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all Powertrain CAN ECUs																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Powertrain CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU Powertrain CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Powertrain CAN .dbc version [] = [			4	ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	4	ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte	M	BCD	5	ECU Powertrain CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Powertrain CAN .dbc version [] = [																										
	4	ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	4	ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte	M	BCD																								
	5	ECU Powertrain CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 1234																											
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU DBC file version is stored. - All Powertrain CAN ECU should support this DID.																											

Specification

3.21 DID 0xF11B ECU Chassis CAN dbc file version

Service	0x22			
DID	0xF11B			
Short Description	ECU Chassis CAN .dbc file version			
Implementation	Application	Flash Boot Loader		
	Read	N/A		
Security Level	Read	Write		
	Level 0	N/A		
Usage	Mandatory for all Chassis CAN ECUs			
Data				
	Data Byte No.	Parameter Name	Cvt	Value
		ECU Chassis CAN .dbc version [] = [		
	4	ECU Chassis CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
	4	ECU Chassis CAN .dbc version -high byte#0- low nibble Byte	M	BCD
	5	ECU Chassis CAN .dbc version high byte#1	M	BCD
		]		
	Parameter:			
	• e.g. 1234			

Specification

3.22 DID 0xF10E ECU Infotainment CAN dbc file version

Service	0x22									
DID	0xF10E									
Short Description	ECU Infotainment CAN .dbc file version									
Implementation	Application	Flash Boot Loader								
	Read	N/A								
Security Level	Read	Write								
	Level 0	N/A								
Usage	Mandatory for all Infotainment CAN ECUs									
Data										
	Data Byte No.	Parameter Name	Cvt	Value						
		ECU Infotainment CAN .dbc version [] = [								
	4	ECU Infotainment CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned						
	4	ECU Infotainment CAN .dbc version -high byte#0- low nibble Byte	M	BCD						
	5	ECU Infotainment CAN .dbc version high byte#1	M	BCD						
		]								
	Parameter:s									
	• e.g. 1234									
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34
	Byte4 high Nibble	Byte4 low Nibble	Byte5							
	1	2	34							
	Note:									
	• Defined by OEM, but written at the supplier side.									
• All byte 0 means no ECU DBC file version is stored.										
• All Infotainment CAN ECU should support this DID.										

Specification

3.23 DID 0xF125 ECU ADAS CAN dbc file version

Service	0x22																											
DID	0xF125																											
Short Description	ECU connected to ADAS CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for ECU's on ADAS CAN																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU ADAS CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU ADAS CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU ADAS CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU ADAS CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU ADAS CAN .dbc version [] = [			4	ECU ADAS CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	4	ECU ADAS CAN .dbc version -high byte#0- low nibble Byte	M	BCD	5	ECU ADAS CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU ADAS CAN .dbc version [] = [																										
	4	ECU ADAS CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	4	ECU ADAS CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	5	ECU ADAS CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 1234																											
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																									
	1	2	34																									
Note:																												
- Defined by OEM, but written at the supplier side. - All byte 0 means no Diagnostic DBC file version is stored. - All ADAS CAN ECU should support this DID.																												

Specification

3.24 DID 0xF126 ECU Diagnostic CAN dbc file version

Service	0x22									
DID	0xF126									
Short Description	ECU connected to Diagnostic CAN .dbc file version									
Implementation	Application	Flash Boot Loader								
	Read	N/A								
Security Level	Read	Write								
	Level 0	N/A								
Usage	Mandatory for ECU's on Diagnostic CAN									
Data										
	Data Byte No.	Parameter Name	Cvt	Value						
		ECU Diag CAN .dbc version [] = [								
	4	ECU Diag CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned						
	4	ECU Diag CAN .dbc version -high byte#0- low nibble Byte	M	BCD						
	5	ECU Diag CAN .dbc version high byte#1	M	BCD						
		]								
	Parameter:									
	• e.g. 1234									
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34
	Byte4 high Nibble	Byte4 low Nibble	Byte5							
	1	2	34							
	Note:									
	• Defined by OEM, but written at the supplier side.									
	• All byte 0 means no Diagnostic DBC file version is stored.									
• All Diagnostic CAN ECU should support this DID.										

Specification

3.25 DID 0xF127 ECU FA CAN dbc file version

Service	0x22																											
DID	0xF127																											
Short Description	ECU connected to FA CAN .dbc file version																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for ECU's on FA CAN (VF35 ICE only)																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU FA CAN .dbc version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU FA CAN .dbc version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU FA CAN .dbc version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU FA CAN .dbc version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU FA CAN .dbc version [] = [			4	ECU FA CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned	4	ECU FA CAN .dbc version -high byte#0- low nibble Byte	M	BCD	5	ECU FA CAN .dbc version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU FA CAN .dbc version [] = [																										
	4	ECU FA CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned																								
	4	ECU FA CAN .dbc version -high byte#0- low nibble Byte	M	BCD																								
	5	ECU FA CAN .dbc version high byte#1	M	BCD																								
		]																										
	Parameter:																											
	e.g. 1234																											
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no FA DBC file version is stored. - All FA CAN ECU should support this DID.																											

Specification

3.26 DID 0xF128 ECU FLEXRAY FIBEX file version

Service	0x22									
DID	0xF128									
Short Description	ECU connected to FLEXRAY FIBEX file version									
Implementation	Application	Flash Boot Loader								
	Read	N/A								
Security Level	Read	Write								
	Level 0	N/A								
Usage	Mandatory for ECU's on FLEXRAY bus (VF35 ICE only)									
Data										
	Data Byte No.	Parameter Name	Cvt	Value						
		ECU Flexray Fibex .dbc version [] = [								
	4	ECU Flexray Fibex file version -high byte#0— High nibble Byte	M	Unsigned						
	4	ECU Flexray Fibex file version -high byte#0- low nibble Byte	M	BCD						
	5	ECU Flexray Fibex file version high byte#1	M	BCD						
		]								
	Parameter:									
	• e.g. 1234									
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34
	Byte4 high Nibble	Byte4 low Nibble	Byte5							
	1	2	34							
	Note:									
	• Defined by OEM, but written at the supplier side.									
	• All byte 0 means no Flexray Fibex file version is stored.									
• All Flexray Fibex FA CAN ECU should support this DID.										

Specification

3.27 DID 0xF10D ECU LIN1 LDF file version

Service	0x22																											
DID	0xF10D																											
Short Description	ECU LIN1 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN1 ECUs																											
Data	<table><thead><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr></thead><tbody><tr><td></td><td>ECU LIN1 .ldf version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU LIN1 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU LIN1 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU LIN1 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></tbody></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN1 .ldf version [] = [			4	ECU LIN1 .ldf version -high byte#0— High nibble Byte	M	Unsigned	4	ECU LIN1 .ldf version -high byte#0- low nibble Byte	M	BCD	5	ECU LIN1 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN1 .ldf version [] = [																										
	4	ECU LIN1 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	4	ECU LIN1 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	5	ECU LIN1 .ldf version high byte#1	M	BCD																								
		]																										
	Parameter:s																											
	e.g. 1234																											
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU LDF file version is stored. - All LIN1 CAN ECU should support this DID.																											

Specification

3.28 DID 0xF11C ECU LIN2 LDF file version

Service	0x22																											
DID	0xF11C																											
Short Description	ECU LIN2 .ldf file version																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all LIN2 ECUs																											
Data	<table><thead><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr></thead><tbody><tr><td></td><td>ECU LIN2 .ldf version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU LIN2 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU LIN2 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU LIN2 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></tbody></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU LIN2 .ldf version [] = [			4	ECU LIN2 .ldf version -high byte#0— High nibble Byte	M	Unsigned	4	ECU LIN2 .ldf version -high byte#0- low nibble Byte	M	BCD	5	ECU LIN2 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU LIN2 .ldf version [] = [																										
	4	ECU LIN2 .ldf version -high byte#0— High nibble Byte	M	Unsigned																								
	4	ECU LIN2 .ldf version -high byte#0- low nibble Byte	M	BCD																								
	5	ECU LIN2 .ldf version high byte#1	M	BCD																								
		]																										
	Parameter:s																											
	e.g. 1234																											
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>				Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																									
	1	2	34																									
	Note:																											
	- Defined by OEM, but written at the supplier side. - All byte 0 means no ECU LDF file version is stored. - All LIN2 CAN ECU should support this DID.																											

Specification

3.29 DID 0xF129 ECU LIN3 LDF file version

Service	0x22																										
DID	0xF129																										
Short Description	ECU LIN3 .ldf file version																										
Implementation	Application	Flash Boot Loader																									
	Read	N/A																									
Security Level	Read	Write																									
	Level 0	N/A																									
Usage	Mandatory for all LIN3 ECUs																										
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU LIN3 .ldf version [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU LIN3 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr><tr><td>4</td><td>ECU LIN3 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr><tr><td>5</td><td>ECU LIN3 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>			Data Byte No.	Parameter Name	Cvt	Value		ECU LIN3 .ldf version [] = [			4	ECU LIN3 .ldf version -high byte#0— High nibble Byte	M	Unsigned	4	ECU LIN3 .ldf version -high byte#0- low nibble Byte	M	BCD	5	ECU LIN3 .ldf version high byte#1	M	BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																							
		ECU LIN3 .ldf version [] = [																									
	4	ECU LIN3 .ldf version -high byte#0— High nibble Byte	M	Unsigned																							
	4	ECU LIN3 .ldf version -high byte#0- low nibble Byte	M	BCD																							
	5	ECU LIN3 .ldf version high byte#1	M	BCD																							
		]																									
	Parameter:																										
	e.g. 1234																										
	<table><tr><td>Byte4 high Nibble</td><td>Byte4 low Nibble</td><td>Byte5</td></tr><tr><td>1</td><td>2</td><td>34</td></tr></table>			Byte4 high Nibble	Byte4 low Nibble	Byte5	1	2	34																		
	Byte4 high Nibble	Byte4 low Nibble	Byte5																								
	1	2	34																								
	Note:																										
	- Defined by OEM, but written at the supplier side. - All byte 0 means no BCM LIN1 LDF file version is stored. - All LIN3 ECU should support this DID.																										

Specification

3.30 DID 0xF10F Service Date

Service	0x22 / 0x2E																										
DID	0xF10F																										
Short Description	The date of last Service																										
Implementation	Application	Flash Boot Loader																									
	Read/Write	N/A																									
Security Level	Read	Write																									
	Level 0	Level 1																									
Usage	Mandatory for all ECU's which support diagnostic																										
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Last Vehicle servicing performed Date[] = [</td><td></td><td></td></tr><tr><td>4</td><td>Service Date - Year</td><td>M</td><td>(00 - 99) BCD</td></tr><tr><td>5</td><td>Service Date - Month</td><td>M</td><td>(01 - 12) BCD</td></tr><tr><td>6</td><td>Service Date - Day</td><td>M</td><td>(01 - 31) BCD</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>			Data Byte No.	Parameter Name	Cvt	Value		Last Vehicle servicing performed Date[] = [			4	Service Date - Year	M	(00 - 99) BCD	5	Service Date - Month	M	(01 - 12) BCD	6	Service Date - Day	M	(01 - 31) BCD		]		
	Data Byte No.	Parameter Name	Cvt	Value																							
		Last Vehicle servicing performed Date[] = [																									
	4	Service Date - Year	M	(00 - 99) BCD																							
	5	Service Date - Month	M	(01 - 12) BCD																							
	6	Service Date - Day	M	(01 - 31) BCD																							
		]																									
	Parameter:																										
	- This date is written by tester tool at the time of a service ID 0x14 (Physical or Functional addressing). - Any other reasons for writing this DID need to be approved by the Vinfast Aftersales Diagnostic and Programming Team																										
	e.g: 2018.01.21																										
	<table><tr><td>Byte4</td><td>Byte5</td><td>Byte6</td></tr><tr><td>18</td><td>01</td><td>21</td></tr></table>			Byte4	Byte5	Byte6	18	01	21																		
	Byte4	Byte5	Byte6																								
	18	01	21																								
Note:																											
The default value is all byte 0x00.																											

Specification

3.31 DID 0xFD01 Supplier Specific Part Number Data Identifier

Service	0x22			
DID	0XFD01			
Short Description	Read Supplier specific part number			
Implementation	Application	Flash Boot Loader		
	Read	N/A		
Security Level	Read	Write		
	Level 0	N/A		
Usage	Optional or User Conditional			
Data	Data Byte No.	Parameter Name	Cvt	Value
		Supplier specific part number [] = [		
	4	Supplier Specific Part Number High Byte#0	M	ASCII
	...		...	...
	4+n-1	Supplier Specific Part Number Low Byte#n	M	ASCII
		]		
	Parameter:			
	It is allowed to use a variable String length. See the component diagnostic information.			
	Note:			
	The supplier specific part number shall be defined and written by the Supplier.			

3.32 DID 0xF192 System Supplier ECU Hardware Number

Service	0x22																											
DID	0XF192																											
Short Description	Read System Supplier hardware part number																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Optional or User Conditional																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Supplier specific hardware number [] = [</td><td></td><td></td></tr><tr><td>4</td><td>Supplier Specific hardware Number High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>4+n-1</td><td>Supplier Specific hardware Number Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Supplier specific hardware number [] = [			4	Supplier Specific hardware Number High Byte#0	M	ASCII	...		...	...	4+n-1	Supplier Specific hardware Number Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Supplier specific hardware number [] = [																										
	4	Supplier Specific hardware Number High Byte#0	M	ASCII																								
	...		...	...																								
	4+n-1	Supplier Specific hardware Number Low Byte#n	M	ASCII																								
		]																										
	Parameter: It is allowed to use a variable String length. See also the component diagnostic information.																											
	Note: The system supplier Hardware part number shall be defined and written by the Supplier.																											

Specification

3.33 DID 0xF194 System Supplier ECU Software Number

-Service	0x22			
DID	0XF194			
Short Description	Read Supplier specific Software number			
Implementation	Application	Flash Boot Loader		
	Read	N/A		
Security Level	Read	Write		
	Level 0	N/A		
Usage	Optional or User Conditional			
Data	Data Byte No.	Parameter Name	Cvt	Value
		Supplier specific Software number [] = [		
	4	Supplier Specific Software Number High Byte#0	M	ASCII
	...		...	...
	4+n-1	Supplier Specific Software Number Low Byte#n	M	ASCII
		]		
	Parameter: - It is allowed to use a variable String length. See also the component diagnostic information. Note: - The Software number shall be defined and written by the Supplier.			

3.34 DID 0xF110 ECU name data identifier

Service	0x22																											
DID	0XF110																											
Short Description	Read ECU Name																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all ECU's which support diagnostic																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>ECU Name [] = [</td><td></td><td></td></tr><tr><td>4</td><td>ECU Name High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td>.....</td><td>...</td><td>...</td></tr><tr><td>18</td><td>ECU Name Low Byte#n</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		ECU Name [] = [			4	ECU Name High Byte#0	M	ASCII	...		...	...	18	ECU Name Low Byte#n	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		ECU Name [] = [																										
	4	ECU Name High Byte#0	M	ASCII																								
	...		...	...																								
	18	ECU Name Low Byte#n	M	ASCII																								
		]																										
	Parameter: - The max. length of the string is 15 Byte. - The values should be aligned left. - Unused Bytes should be filled with space (0x20)																											
	Note: The ECU name shall be defined by the OEM, but should be written by the Supplier.																											

Specification

3.35 DID 0xF120 Global Time

Service	0x22		
DID	0XF120		
Short Description	Global Time		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic		
Data			

Specification

3.36 DID 0xF121 Odometer

Service	0x22								
DID	0XF121								
Short Description	Odometer								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>3</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	3	M	Unsigned
	Size (Bytes)	Cvt	Type						
	3	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = IC_MILEAGE - Resolution is 1km/bit. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

3.37 DID 0xF122 Vehicle Supply Voltage

Service	0x22								
DID	0XF122								
Short Description	Vehicle Supply Voltage								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = BCM_VOLTAGE - Resolution is 0.1V/bit - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

Specification

3.38 DID 0xF123 Ignition status

Service	0x22							
DID	0XF123							
Short Description	Ignition status							
Implementation	Application	Flash Boot Loader						
	Read	N/A						
Security Level	Read	Write						
	Level 0	N/A						
Usage	Mandatory for all ECU's which support diagnostic							
Data	<table border="1"> <thead> <tr> <th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr> </thead> <tbody> <tr> <td>1</td><td>M</td><td>Packeted</td></tr> </tbody> </table> Parameter: 0x00=OFF 0x01=ACC 0x02=ON 0x03=START 0x04~0x06=reserved 0x07=invalid - Reference CAN signal message = BCM_CLAMP_STAT - The values should be aligned left. - Unused Bytes should be filled with space (0x00) Note: - The ECU name shall be defined by the OEM, but should be written by the Supplier.		Size (Bytes)	Cvt	Type	1	M	Packeted
Size (Bytes)	Cvt	Type						
1	M	Packeted						

3.39 DID 0xF18C ECUSerialNumberDataIdentifier

Service	0x22							
DID	0XF18C							
Short Description	ECUSerialNumberDataIdentifier							
Implementation	Application	Flash Boot Loader						
	Read	Read / Write (once)						
Security Level	Read	Write						
	Level 0	Level 3						
Usage	Mandatory for all ECU's which support diagnostic							
Data	<table border="1"> <thead> <tr> <th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr> </thead> <tbody> <tr> <td>16</td><td>M</td><td>ASCII</td></tr> </tbody> </table> Parameter: - The values should be aligned right. - Unused Bytes should be filled with space (0x00) Note: - The ECU name shall be defined by the OEM, but should be written by the Supplier.		Size (Bytes)	Cvt	Type	16	M	ASCII
Size (Bytes)	Cvt	Type						
16	M	ASCII						

Specification

3.40 DID 0xDD01 Cabin temperature

Service	0x22								
DID	0XDD01								
Short Description	Cabin temperature								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = CCU_INTERIORTEMP - Supported based on ECU location. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

3.41 DID 0xDD02 Outside temperature

Service	0x22								
DID	0XDD02								
Short Description	Outside temperature								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = TEMP_EX_UNFILT - Supported based on ECU location - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

Specification

3.42 DID 0xDD03 ECU temperature

Service	0x22								
DID	0XDD03								
Short Description	ECU temperature								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

3.43 DID 0xDD04 ECU Voltage

Service	0x22								
DID	0XDD04								
Short Description	ECU Voltage								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data									
	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
- Resolution is 0.1V/bit - The values should be aligned left. - Unused Bytes should be filled with space (0x00)									
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

Specification

3.44 DID 0xDD05 Motor RPM

Service	0x22								
DID	0XDD05								
Short Description	Motor RPM								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>2</td><td>O/M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	2	O/M	Unsigned
	Size (Bytes)	Cvt	Type						
	2	O/M	Unsigned						
	Parameter:								
	- Byte 1: Reference CAN signal message = EDS_SPD_FR - Byte 2: Reference CAN signal message = EDS_SPD_RR - Supported only on BEV vehicles. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
	Note:								
	- The ECU name shall be defined by the OEM, but should be written by the Supplier. - Resolution factor is 80/bit = 20.400 with max value of 16.000RPM								

3.45 DID 0xDD06 Engine status

Service	0x22		
DID	0XDD06		
Short Description	Engine status		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic (ICE only)		
Data			
	Size (Bytes)	Cvt	Type
	1	O/M	Bit Mapped
	Parameter:		
	- Supported only on ICE vehicles. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)		
Note:			
	The ECU name shall be defined by the OEM, but should be written by the Supplier.		

Specification

3.46 DID 0xDD07 GPS Time

Service	0x22								
DID	0XDD07								
Short Description	GPS Time								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>4</td><td>O/M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	4	O/M	Unsigned
	Size (Bytes)	Cvt	Type						
	4	O/M	Unsigned						
	Parameter:								
	- Reference CAN signal message = MHU_DATE_TIME - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
The ECU name shall be defined by the OEM, but should be written by the Supplier.									

3.47 DID 0xDD08 Engine RPM

Service	0x22								
DID	0XDD08								
Short Description	Engine RPM								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic (ICE only)								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>O/M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	O/M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	O/M	Unsigned						
	Parameter:								
	- Supported only on ICE vehicles. - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
	Note:								
	- The ECU name shall be defined by the OEM, but should be written by the Supplier. - Resolution is 50/bit = 12750 with max value of 10000RPM.								

Specification

3.48 DID 0xDD09 Vehicle speed

Service	0x22								
DID	0XDD09								
Short Description	Vehicle Speed								
Implementation	Application	Flash Boot Loader							
	Read	N/A							
Security Level	Read	Write							
	Level 0	N/A							
Usage	Mandatory for all ECU's which support diagnostic								
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>M</td><td>Unsigned</td></tr></table>			Size (Bytes)	Cvt	Type	1	M	Unsigned
	Size (Bytes)	Cvt	Type						
	1	M	Unsigned						
	Parameter:								
	- Reference CAN signal message = VEHICLESPD - The values should be aligned left. - Unused Bytes should be filled with space (0x00)								
Note:									
- The ECU name shall be defined by the OEM, but should be written by the Supplier. - Resolution is 1/bit = 255km/h									

3.49 DID 0xD00D Vehicle Network Communication Bus State

Service	0x22		
DID	0XD00D		
Short Description	Vehicle Network Communication Bus State		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Optional for all ECU's which support diagnostic		
Data			

Specification

3.50 DID 0xD020 Internal Fail Code

Service	0x22																																																																													
DID	0XD020																																																																													
Short Description	Internal Fail Code																																																																													
Implementation	Application			Flash Boot Loader																																																																										
	Read			N/A																																																																										
Security Level	Read			Write																																																																										
	Level 0			N/A																																																																										
Usage	Optional for all ECU's which support diagnostic																																																																													
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>105</td><td>O</td><td>Packeted</td></tr></table>					Size (Bytes)	Cvt	Type	105	O	Packeted																																																																			
	Size (Bytes)	Cvt	Type																																																																											
	105	O	Packeted																																																																											
	<table><tr><td colspan="2">1 byte state encoded</td><td></td><td>8 bytes Numeric</td><td></td><td>unsigned numeric</td></tr><tr><td>State</td><td>1 byte</td><td></td><td>1/2byte Fault Count max count 8 (8 byte)</td><td>Fault count</td><td>Last Time occurrence (96 bytes)</td></tr><tr><td>0000</td><td>Failure description</td><td>CAN transceiver fault</td><td>Number of counts per fault</td><td>max count 8</td><td>yymmddhhmm</td></tr><tr><td>0001</td><td>1</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>0010</td><td>2</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>0011</td><td>3</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>0100</td><td>4</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>0101</td><td>5</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>0110</td><td>6</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>0111</td><td>7</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>1000</td><td>8</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr><tr><td>1001</td><td>9</td><td>name of device failed</td><td>Number of counts per fault</td><td>max count 8</td><td>6</td></tr></table>						1 byte state encoded			8 bytes Numeric		unsigned numeric	State	1 byte		1/2byte Fault Count max count 8 (8 byte)	Fault count	Last Time occurrence (96 bytes)	0000	Failure description	CAN transceiver fault	Number of counts per fault	max count 8	yymmddhhmm	0001	1	name of device failed	Number of counts per fault	max count 8	6	0010	2	name of device failed	Number of counts per fault	max count 8	6	0011	3	name of device failed	Number of counts per fault	max count 8	6	0100	4	name of device failed	Number of counts per fault	max count 8	6	0101	5	name of device failed	Number of counts per fault	max count 8	6	0110	6	name of device failed	Number of counts per fault	max count 8	6	0111	7	name of device failed	Number of counts per fault	max count 8	6	1000	8	name of device failed	Number of counts per fault	max count 8	6	1001	9	name of device failed	Number of counts per fault	max count 8	6
	1 byte state encoded			8 bytes Numeric		unsigned numeric																																																																								
	State	1 byte		1/2byte Fault Count max count 8 (8 byte)	Fault count	Last Time occurrence (96 bytes)																																																																								
	0000	Failure description	CAN transceiver fault	Number of counts per fault	max count 8	yymmddhhmm																																																																								
	0001	1	name of device failed	Number of counts per fault	max count 8	6																																																																								
	0010	2	name of device failed	Number of counts per fault	max count 8	6																																																																								
	0011	3	name of device failed	Number of counts per fault	max count 8	6																																																																								
	0100	4	name of device failed	Number of counts per fault	max count 8	6																																																																								
	0101	5	name of device failed	Number of counts per fault	max count 8	6																																																																								
	0110	6	name of device failed	Number of counts per fault	max count 8	6																																																																								
	0111	7	name of device failed	Number of counts per fault	max count 8	6																																																																								
1000	8	name of device failed	Number of counts per fault	max count 8	6																																																																									
1001	9	name of device failed	Number of counts per fault	max count 8	6																																																																									

Specification

	1010	10	name of device failed	Number of counts per fault	max count 8	6	
	1011	11	name of device failed	Number of counts per fault	max count 8	6	
	1100	12	name of device failed	Number of counts per fault	max count 8	6	
	1101	13	name of device failed	Number of counts per fault	max count 8	6	
	1110	14	name of device failed	Number of counts per fault	max count 8	6	
	1111	15	name of device failed	Number of counts per fault	max count 8	6	
Parameter: - The values should be aligned left. - Unused Bytes should be filled with space (0x00) Note: - The ECU name shall be defined by the OEM, but should be written by the Supplier.							

3.51 DID 0x4884 Internal Fault Data

Service	0x22		
DID	0X4884		
Short Description	Internal Fault Data		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Optional for all ECU's which support diagnostic		
Data			
	Size (Bytes)	Cvt	Type
	32	O	HEX
	Parameter:		
	- The values should be aligned left. - Unused Bytes should be filled with space (0x00)		
Note:			
	The ECU name shall be defined by the OEM, but should be written by the Supplier.		

Specification

3.52 DID 0xF800 Vehicle Identification Number 2 (VIN2)

Service	0x22																											
DID	0XF800																											
Short Description	VIN2																											
Implementation	Application		Flash Boot Loader																									
	Read		N/A																									
Security Level	Read		Write																									
	Level 0		N/A																									
Usage	Mandatory for all ECU's which support diagnostic																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>4</td><td>Vehicle Identification Number High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>20</td><td>Vehicle Identification Number Low Byte#16</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			4	Vehicle Identification Number High Byte#0	M	ASCII	...				20	Vehicle Identification Number Low Byte#16	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	4	Vehicle Identification Number High Byte#0	M	ASCII																								
	...																											
	20	Vehicle Identification Number Low Byte#16	M	ASCII																								
		]																										
	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																											
	Note: - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later processes. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER																											

3.53 DID 0xF801 Vehicle Identification Number 3 (VIN3)

Service	0x22																											
DID	0XF801																											
Short Description	VIN3																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all ECU's which support diagnostic																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>4</td><td>Vehicle Identification Number High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>20</td><td>Vehicle Identification Number Low Byte#16</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			4	Vehicle Identification Number High Byte#0	M	ASCII	...				20	Vehicle Identification Number Low Byte#16	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	4	Vehicle Identification Number High Byte#0	M	ASCII																								
	...																											
	20	Vehicle Identification Number Low Byte#16	M	ASCII																								
		]																										

Specification

	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212 Note: - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later processes. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER
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3.54 DID 0xF802 Vehicle Identification Number 4 (VIN4)

Service	0x22																											
DID	0XF802																											
Short Description	VIN4																											
Implementation	Application	Flash Boot Loader																										
	Read	N/A																										
Security Level	Read	Write																										
	Level 0	N/A																										
Usage	Mandatory for all ECU's which support diagnostic																											
Data	<table><tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr><tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr><tr><td>4</td><td>Vehicle Identification Number High Byte#0</td><td>M</td><td>ASCII</td></tr><tr><td>...</td><td></td><td></td><td></td></tr><tr><td>20</td><td>Vehicle Identification Number Low Byte#16</td><td>M</td><td>ASCII</td></tr><tr><td></td><td>]</td><td></td><td></td></tr></table>				Data Byte No.	Parameter Name	Cvt	Value		Vehicle Identification Number [] = [			4	Vehicle Identification Number High Byte#0	M	ASCII	...				20	Vehicle Identification Number Low Byte#16	M	ASCII		]		
	Data Byte No.	Parameter Name	Cvt	Value																								
		Vehicle Identification Number [] = [																										
	4	Vehicle Identification Number High Byte#0	M	ASCII																								
	...																											
	20	Vehicle Identification Number Low Byte#16	M	ASCII																								
		]																										
	Parameter: - Vehicle Identification Number format: XXXXXXXXXXXXXXXXXXXX - e.g.: WBCN1233VVV333212																											
	Note: - The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later processes. - The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. - All Bytes 0x00 means no Vehicle Identification Number is stored. - Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER																											

Specification

3.55 DID 0xF141 Vehicle Manufacturer ECU Hardware Part Number Revision Level

Service	0x22					
DID	0xF141					
Short Description	Hardware Number Revision Level of the ECU defined by the OEM.					
Implementation	Application	Flash Boot Loader				
	Mandatory Read(physical and functional addressing)	Read				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Hardware Part Number Revision Level [] = [				
	4	ECU Hardware Number Revision Level	M	BCD		
		]				
	Parameter:					
	- ECU Hardware Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Hardware Part Number found in DID 0xF191					
	e.g.: EEP12345678-01					
	<table><tr><td>Byte4</td></tr><tr><td>01</td></tr></table>				Byte4	01
	Byte4					
01						
Note:						
- The OEM defines Vehicle Manufacture ECU Hardware Number Revision Level. - The Hardware Number Revision Level must be written by the Supplier. - Byte4 0x00 means no Vehicle Manufacture ECU Hardware Number Revision Level is stored.						

Specification

3.56 DID 0xF142 Vehicle Manufacturer ECU Calibration Part Number Revision Level

Service	0x22					
DID	0xF142					
Short Description	Calibration Software Number Revision Level of the ECU defined by the OEM					
Implementation	Application	Flash Boot Loader				
	Mandatory Read	Updated during software download Not readable in Boot Loader				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory for ECUs where Calibration Part Number is to be reported separately					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		Calibration Software Part Number Revision Level [] = [				
	4	ECU Calibration Software Number Release Level	M	BCD		
		]				
	Parameter:					
	- ECU Calibration Software Part Number Release Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Calibration Software Part Number found in DID 0xF102					
	e.g.: SOW11223344-01					
	<table><tr><td>Byte4</td></tr><tr><td>01</td></tr></table>				Byte4	01
	Byte4					
01						
Note:						
- The Calibration Software Number Revision Level is defined by the OEM, and shall be embedded in the ECU Calibration Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Vehicle Manufacture ECU Calibration Software Number Revision Level is stored.						

Specification

3.57 DID 0xF143 Vehicle Manufacturer Purchase Part Number Revision Level

Service	0x22					
DID	0xF143					
Short Description	Purchase Part Number Revision Level of the ECU defined by the OEM					
Implementation	Application		Flash Boot Loader			
	Mandatory Read(physical and functional addressing)		Updated during software download Not readable in Boot Loader			
Security Level	Read		Write			
	Level 0		N/A			
Usage	Mandatory					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Part Number Revision Level [] = [				
	4	ECU Number Revision Level	M	BCD		
Parameter: - ECU Purchase Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Number found in DID 0xF103 e.g. EEP12345678-01 <table><tr><td>Byte4</td></tr><tr><td>01</td></tr></table> Note: - The ECU Purchase Part Number Revision Level is defined by the OEM and must be written at Supplier manufacturing process. - Byte4 0x00 means no Vehicle Manufacture ECU Hardware Revision Level is stored.					Byte4	01
Byte4						
01						

Specification

3.58 DID 0xF144 Vehicle Manufacturer ECU Application Software Part Number Revision Level

Service	0x22					
DID	0xF144					
Short Description	Application Software Part Number Revision Level defined by the OEM					
Implementation	Application	Flash Boot Loader				
	Read	Updated during software download Not readable in Boot Loader				
Security Level	Read	Write				
	Level 0	N/A				
Usage	Mandatory for ECUs with flashing function and whose application software (ASW) packages can be programmed separately from basic software (BSW), else use DID 0xF188					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Application Software Part Number Revision Level [] = [				
	4	ECU Application Software Number Revision Level	M	BCD		
		]				
	Parameter:					
	- Application Software Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of Application Software Number found in DID 0xF104					
	e.g. SOW12345678-01					
	<table><tr><td>Byte4</td></tr><tr><td>01</td></tr></table>				Byte4	01
	Byte4					
01						
Note:						
- The Application Software Number Revision Level is defined by the OEM, and shall be embedded in the ECU Application Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Application Software Number Revision Level is stored.						

Specification

3.59 DID 0xF145 Vehicle Manufacturer ECU Basic Software Part Number Revision Level

Service	0x22					
DID	0xF145					
Short Description	Basic Software Part Number Revision Level defined by the OEM					
Implementation	Application		Flash Boot Loader			
	Read		Updated during software download Not readable in Boot Loader			
Security Level	Read		Write			
	Level 0		N/A			
Usage	Mandatory for ECUs which with flashing function and whose basic software can be reprogrammed separately					
Data	Data Byte No.	Parameter Name	Cvt	Value		
		ECU Basic Software Part Number Revision Level [] = [				
	4	ECU Basic Software Number Revision Level	M	BCD		
	Parameter:					
	- Basic Software Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of Basic Software Number found in DID 0xF105					
	e.g. SOW12345678-01					
	<table><tr><td>Byte4</td></tr><tr><td>01</td></tr></table>				Byte4	01
	Byte4					
01						
Note:						
- The Basic Software Part Number Revision Level is defined by the OEM, and shall be embedded in the ECU Basic Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Basic Software Part Number Revision Level is stored.						

Specification

3.60 DID 0xF146 Vehicle Manufacturer ECU HMI Software Part Number Revision Level

Service	0x22			
DID	0xF146			
Short Description	HMI Software Part Number Revision Level defined by the OEM			
Implementation	Application		Flash Boot Loader	
	Read		Updated during software download Not readable in Boot Loader	
Security Level	Read		Write	
	Level 0		N/A	
Usage	Mandatory for ECUs with HMI software module and can be reprogrammed.			
Data	Data Byte No.	Parameter Name	Cvt	Value
		ECU HMI Software Part Number Revision Level [] = []		
	4	ECU HMI Software Number Revision Level	M	BCD
	Parameter:			
	- HMI Software Part Number Revision Level from BOM/Teamcenter. - This DID represents the Revision Level of HMI Software Number found in DID 0xF106			
	e.g. SOW12345678-01			
	Byte4 01			
	Note:			
- This HMI Software Part Number Revision Level is defined by the OEM, and shall be embedded in the ECU HMI Software file. - This DID will be updated during the reprogramming process. - Byte4 0x00 means no HMI Software Number Revision Level is stored.				

Specification

3.61 DID 0xF148 Vehicle Manufacturer ECU Software Part Number Revision Level

Service	0x22						
DID	0xF148						
Short Description	Software Part Number Revision Level of the ECU defined by the OEM.						
Implementation	Application		Flash Boot Loader				
	Mandatory Read(physical and functional addressing)		Updated during software download Not readable in Boot Loader				
Security Level	Read		Write				
	Level 0		N/A				
Usage	Mandatory for all ECUs.						
Data	Data Byte No.	Parameter Name	Cvt	Value			
		ECU Software Part Number Revision Level [] = [					
	4	ECU Software Number Revision Level	M	BCD			
		]					
Parameter: - ECU Software Part Number Revision Level from BOM/Teamcenter: - This DID represents the Revision Level of ECU Software Part Number found in DID 0xF188 e.g.: SOW12345678-01 <table><tr><td>Byte4</td></tr><tr><td>01</td></tr></table>						Byte4	01
Byte4							
01							
Note: - The OEM defines Vehicle Manufacture ECU Software Number Revision Level, and it shall be embedded in the ECU Software file. - This DID is updated during the reprogramming process. - Byte4 0x00 means no Vehicle Manufacture ECU Software Revision Number is stored.							

Specification

3.62 DID 0x016C HV Status and Charging Status

Service	0x22		
DID	0X016C		
Short Description	HV Status and Charging Status		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic		
Data	Size (Bytes)	Cvt	Type
	1	M	Unsigned
	Parameters:		
	1) HV Status:		
	▪ 0x00= HV OFF		
	▪ 0x01=HV ON		
	2) HV Charging Status:		
	▪ 0x00= Charging OFF		
	▪ 0x01= Charging ON		
	• The values should be aligned left.		
• Unused Bytes should be filled with space (0x00)			
Note:			
• The ECU name shall be defined by the OEM, but should be written by the Supplier.			

3.63 DID 0xD06B Signed Software Consistency Check

Service	0x22		
DID	0XD06B		
Short Description	Signed Software Consistency Check		
Implementation	Application	Flash Boot Loader	
	Read	N/A	
Security Level	Read	Write	
	Level 0	N/A	
Usage	Mandatory for all ECU's which support diagnostic		
Data	Size (Bytes)	Cvt	Type
	1	M	State encoded
	Parameters: 00 = Consistency not checked		
	01 = Consistency checked & ok		
	02 = Signature doesn't match		
	03 = Error and detail of the error		
	- The values should be aligned left. - Unused Bytes should be filled with space (0x00)		
	Note:		
	The ECU name shall be defined by the OEM, but should be written by the Supplier.		

Specification

3.64 DID 0xD05B Software Status Check

Service	0x22				
DID	0xD05B				
Short Description	Software Part Number Status Check				
Implementation	Application	Flash Boot Loader			
	Mandatory Read(physical and functional addressing)	Updated during software download Not readable in Boot Loader			
Security Level	Read	Write			
	Level 0	N/A			
Usage	Mandatory for all ECUs				
Data	Data Byte No.	Parameter Name	Cvt	Value	
		ECU Software Part Number [] = [			
	4	ECU Software Number Prefix Byte#0	M	ASCII	
	5	ECU Software Number Prefix Byte#1	M	ASCII	
	6	ECU Software Number Prefix Byte#2	M	ASCII	
	7	ECU Software Number Byte#0	M	BCD	
	8	ECU Software Number Byte#1	M	BCD	
	9	ECU Software Number Byte#2	M	BCD	
	10	ECU Software Number Byte#3	M	BCD	
	11	ECU Software Number Byte#4	M	State Encoded	
	Parameter:				
	- Byte 4 to Byte 10 = ECU Software Part Number from BOM/Teamcenter - Byte 11: 1 = accepted; 0 = rejected				
	e.g.: SOW1234567800				

Specification

4 Standards

4.1 Additional Documents

Sr. No	Name	Version
[1]	Basic_Diagnostic_Specification	latest
[2]	Flash_Specification	latest
[3]	Coding_Specification	latest
[4]	ECU_Identifier_Specification	latest

Table 4: Referenced Documents

4.2 Norms and Standards

Ref. No	Document Name
ISO14229	Road vehicles — Unified diagnostic services (UDS)
ISO15031	Road vehicles — Communication between vehicle and external equipment for emissions- related Diagnostic
ISO15765	Road vehicles — Diagnostic on Controller Area Networks (CAN)
ISO17987	Road vehicles — Local Interconnect Network LIN
ISO22900	Road vehicles — Modular vehicle communication interface
ISO22901	Road vehicles — Open diagnostic data exchange (ODX)
SAE J1979	E/E Diagnostic Test Modes
SAE J2012	Diagnostic Trouble Code Definitions

Table 5 : Normative references

This Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

5 Lists

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5.2 List of Figures

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